

SML

Assignment - 1

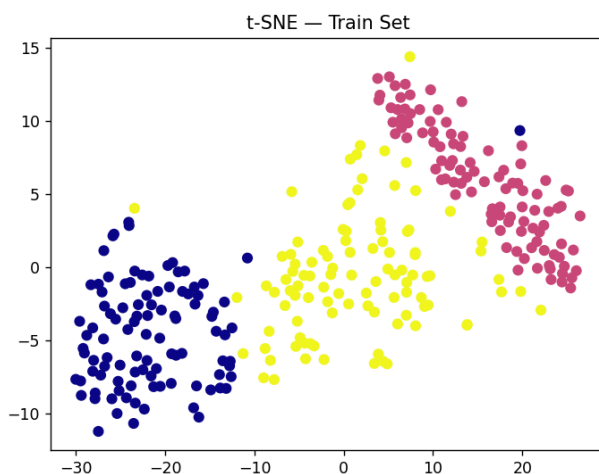
Loading MNIST Dataset

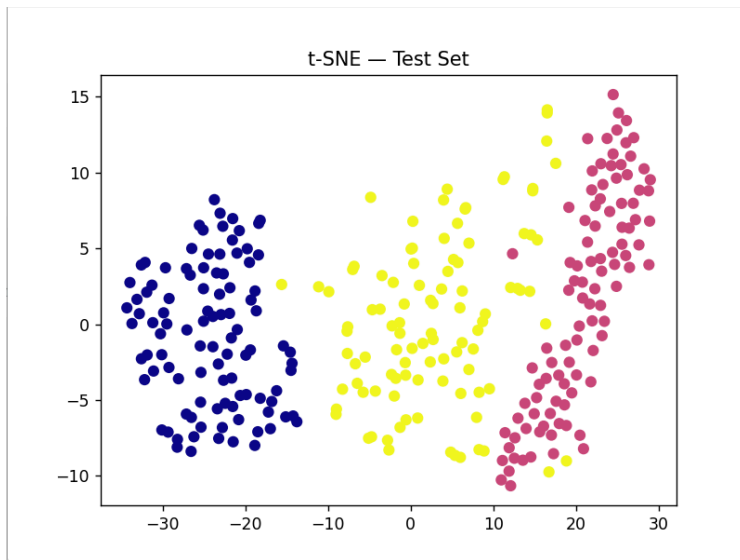
The MNIST handwritten digit dataset was downloaded locally and loaded into Python using the `idx2numpy` library, which converts the binary `IDX` files into NumPy arrays for numerical processing. After loading, the dataset was filtered to include only the digit classes 0, 1, and 2, as required for the experiment.

To create a balanced dataset, 100 samples were randomly selected from each class for both the training and test sets. This resulted in datasets containing 300 samples each, ensuring equal representation of all classes.

Each image, originally stored as a 28×28 matrix of pixel intensities, was converted into a one-dimensional feature vector by stacking its pixel values. The pixel values were then normalized to the range 0 to 1 to improve numerical stability during statistical computations.

t-SNE Plots





The resulting 2D visualization shows how samples from digits 0, 1, and 2 cluster together. Points belonging to the same digit tend to form compact groups, while different digits appear separated. This visual separation supports the effectiveness of statistical classifiers such as LDA and QDA, as it indicates that the classes are distinguishable in feature space.

Maximum Likelihood Estimation (MLE)

Mean formula

$$\mu_c = \frac{1}{N} \sum_{i=1}^N x_i$$

Covariance formula

$$\Sigma_c = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_c)(x_i - \mu_c)^T$$

Gaussian formula

$$P(x \mid y = c) = \frac{1}{(2\pi)^{d/2} |\Sigma_c|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) \right)$$

Maximum Likelihood Estimation was used to estimate the statistical parameters of each digit class that are mean and the covariance matrix. Each class was modeled as a multivariate Gaussian distribution using the training data. The mean vector was computed as the average of all feature vectors in a class, representing its central tendency. The covariance matrix was calculated to capture how features vary together within the class.

These estimated parameters define the Gaussian model for each digit and are later used by LDA and QDA to compute discriminant scores. MLE provides a principled way to learn class statistics directly from the data, forming the foundation of the classification.

LDA & QDA

LDA

Linear Discriminant Analysis is a Gaussian-based classifier that assumes all classes share a common covariance matrix but have different mean vectors. Using these estimates, a discriminant function is computed for each class, and the class with the highest score is selected as the prediction. Because the covariance is shared, LDA produces linear decision boundaries and provides a stable classification method. The equation is as follows:

$$g_c(x) = \mu_c^T \Sigma^{-1} x - \frac{1}{2} \mu_c^T \Sigma^{-1} \mu_c$$

QDA

Quadratic Discriminant Analysis also models each class using a Gaussian distribution, but allows each class to have its own covariance matrix. This makes the classifier more flexible in capturing class-specific variations. The predicted label is the class with the highest discriminant score, resulting in quadratic decision boundaries. The equation is as follows:

$$g_c(x) = -\frac{1}{2} \ln |\Sigma_c| - \frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c)$$

Classification Summary

The classification pipeline was evaluated using accuracy on the test dataset containing digits 0, 1, and 2. Both LDA and QDA produced high accuracy, indicating that the classes are well separated in feature space like we saw in the t-sne plots.

LDA achieved strong performance using a shared covariance assumption. The covariance was calculated as the average of the covariance matrix of the three classes, resulting in linear decision boundaries.

The discriminant values computed for a sample test point show how each classifier assigns scores to all classes before selecting the final prediction. Higher scores indicate a better match between the sample and the class model.

Overall, the results demonstrate that Gaussian-based discriminant analysis is effective for distinguishing between the selected digit classes.

Also, we could see that QDA performs better than LDA in terms of accuracy.

```
===== Classification Summary =====  
  
LDA -> Correct: 243/300  
LDA accuracy: 0.81 (81.0% Correct)  
  
QDA -> Correct: 298/300  
QDA accuracy: 0.993 (99.33% Correct)  
  
Enter any index for LDA & QDA values(0-299): 150  
  
Random index: 150  
True class: 1  
  
--- LDA scores ---  
digit 0: -5206.44  
digit 1: 12935.31  
digit 2: 4962.2  
LDA value = 12935.31  
  
--- QDA scores ---  
digit 0: -1089354.83  
digit 1: -40041.7  
digit 2: -555384.46  
QDA max score: -40041.7
```

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